

Homework 3

Release Date: 1404/8/18

Deadline: 1404/8/24

1 Autocorrelation of Morlet Wavelet (15 pts)

The signal $x(t) = e^{-t^2} \cos(2\pi f_0 t)$ known as real-valued Morlet wavelet function is shown as bellow.

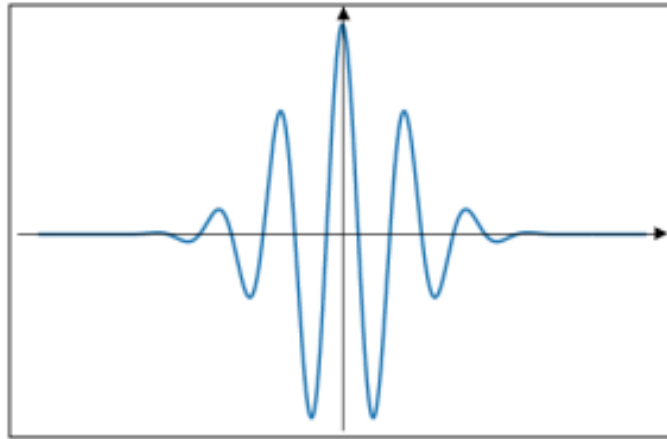


Figure 1: Real-valued Morlet Wavelet.

1. Prove that $x(t)$ is an energy signal.
2. Determine the Fourier transform of this signal.
3. Determine the spectral density function of this signal.
4. Determine the autocorrelation function of this signal.

2 Energy Spectral Density (15 pts)

The voltage $v(t) = Ate^{-\alpha t}u(t)$ is developed across a R ohm resistor, ($\alpha > 0$).

1. Calculate the total energy dissipated in the resistor.
2. What fraction of the energy is contained within a low pass bandwidth of W Hz?
3. What fraction of the energy is contained within a bandwidth of W Hz with a center frequency of f_c Hz?

3 Energy and Power Spectral Density Calculation (20 pts)

Determine whether these signals are energy-type or power-type. In each case, find the energy spectral density and the energy content or the power spectral density and the power content of the signal.

1. $x_1(t) = e^{-\alpha|t|} \sin(\beta t)$, $\alpha, \beta > 0$.
2. $x_2(t) = \text{sinc}^2(3t)$.
3. $x_3(t) = \sum_{n=-\infty}^{\infty} \Lambda(t - 3n)$.
4. $x_4(t) = 2u(-t)$.
5. $x_5(t) = |\cos(2\pi f_0 t)|$.
6. $x_6(t) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{t^2}{2\sigma^2}}$.

4 Double Side-Band Modulation (15 pts)

The message signal $m(t) = 2\cos(300t) + 3\sin(600t + \frac{\pi}{2})$ modulates the carrier signal $c(t) = A\cos(6000\pi t)$, using DSB amplitude modulation. Find the time domain and frequency domain representation of the modulated signal and plot the spectrum (Fourier transform) of the modulated signal. What is the power content of the modulated signal?

5 Amplitude Modulation (15 pts)

An AM signal has the form

$$u(t) = [20 + 2\cos(5000\pi t) + 10\cos(10000\pi t)] \cos(2\pi f_c t)$$

, where $f_c = 20^5$ Hz.

1. Sketch the (voltage) spectrum of $u(t)$.
2. Determine the power in each of the frequency components.
3. Determine the modulation index.
4. Determine the power in the sidebands, the total power, and the ratio of the sidebands power to the total power.

6 A Simple Scrambler System (20 pts)

The Fourier transform of the message signal $m(t)$ is shown in Fig. 2. This signal is applied to the system illustrated in Fig. 3.

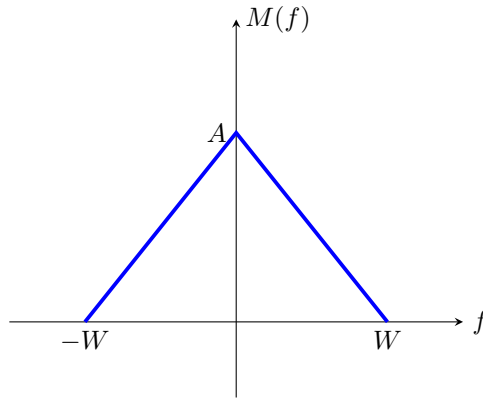


Figure 2: Fourier Transform of $m(t)$.

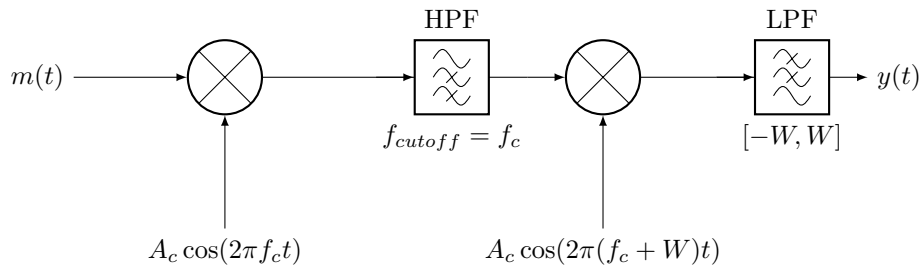


Figure 3: Simple scrambler.

1. Plot $Y(f)$, the Fourier transform of $y(t)$.
2. Show that if $y(t)$ is transmitted, the receiver can pass it through a replica of the system shown in Fig. 3 to obtain $m(t)$ back. This means that this system can be used as a simple scrambler to enhance communication privacy.

Submission

Please submit your answers in a .pdf file named `CS_HW3_studentID.pdf`, replacing `studentID` with your actual student ID.